Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Software engineer pursuing an MSCS with 2+ years building **Python/React internal tools**, SQL-backed services, CI/CD automation, AI-assisted workflows, documentation, and production support systems.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Built **internal platform workflows** for 500+ users, translating stakeholder requests into APIs, validation logic, monitored jobs, technical documentation, release notes, and maintainable self-serve delivery through reviewed changes.
- Improved **SQL-backed service reliability** by tuning indexed queries, background jobs, and CI/CD releases, reducing dashboard latency from **1.2s to 350ms** with traceable handoffs, production reviews, and support notes.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Developed **React dashboards and Flask services** handling 10,000+ daily requests, implementing file ingestion, automated checks, AI-assisted workflows, documented fixes, and sprint-ready delivery with stakeholder feedback loops.
- Partnered with users to debug **business-critical workflow defects** by reproducing OCR, persistence, and execution failures, documenting regression cases, reviewing fixes, and cutting feedback turnaround by **60%** reliably.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Engineered **Python API and ETL services** using SQL, AWS Glue, and S3, improving response time by **30%** across integrations, release reviews, operational handoffs, troubleshooting, supportability, and documentation.
- Maintained **20+ automated test suites** with Jenkins, PyTest, and Mockito across validation flows, increasing CI reliability by **68%** while reducing release regressions during deployments, reviews, and checks.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Built **full-stack business modules** with REST APIs, WebSockets, and SQL/NoSQL persistence, supporting 2M+ synchronized records across dashboards, workflows, integrations, reviews, releases, debugging, validation, monitoring, support, and maintenance.
- Automated **Docker-based build and test flows** with Jenkins, cutting release time by **50%** while improving deployment consistency, troubleshooting documentation, validation notes, and shared-environment handoffs across staging workflows.

Pandit Deendayal Energy University **India**
Data Analyst - Research Fellowship *May 2023 – August 2023*

- Built a **malicious-URL detection pipeline** over 650K+ URLs with 21 behavioral indicators, combining statistical feature engineering, model evaluation, validation reports, threat patterns, reproducibility, and security-focused error analysis.
- Achieved **96%+ detection accuracy** through systematic validation, feature comparison, and model diagnostics, documenting repeatable workflows for large-scale security data review, reproducible experiment handoffs, audits, reviews, and reporting.

Skills

Languages: Python, Go/Golang, JavaScript, TypeScript, SQL, Java, C++, Bash
Frontend: React, Next.js, dashboards, self-serve tools, responsive interfaces, stakeholder workflows
Backend: Flask, FastAPI, REST APIs, service design, integrations, validation, documentation, production support
Infra/Data: PostgreSQL, MySQL, MongoDB, AWS, Azure, Docker, Kubernetes, CI/CD, Jenkins, GitHub Actions
AI/Security: AI-assisted coding, LLM workflows, prompt evaluation, automated agents exposure, security data analysis, monitoring

Projects

Chat2SQL Code | *Java, TypeScript, SQL, Docker*

- Built an **AI-powered SQL workspace** supporting 10+ databases, accelerating query authoring, schema exploration, and database management through Java services, TypeScript interfaces, Docker workflows, Git collaboration, and testing.

Big-Brain Live | *FAISS, PostgreSQL, GCP, LLM APIs*

- Engineered a **cloud retrieval platform** indexing 50K+ documents with FAISS, PostgreSQL, and GCP deployment, reducing retrieval latency by **40%** through optimized ingestion, indexing, caching, and search workflows.

FinNexus Live | *Next.js, TypeScript, MongoDB, Plaid*

- Built a **secure operational dashboard** supporting 50K+ users and 1M+ transactions with token-based access, real-time updates, monitoring, CI/CD deployments, reporting, analytics, audit trails, metrics workflows, and alerts.